Local Sleep Replaces the Sleep Night: Use-Dependent Unit Rotation and Isolated Micro-Batch Replay for Continual Learning Without Offline Phases

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Abstract

Biological brains consolidate memories both in dedicated offline states (sleep) and, as recent causal evidence shows, through *local sleep*: brief, use-dependent off-periods of individual cortical circuits during wakefulness. In contrast, replay-based continual learning in artificial neural networks almost universally relies on offline consolidation phases or on interleaving replayed samples with the input stream. We ask whether a network trained end-to-end by local, biologically constrained learning rules can consolidate *during inference*, with no offline phase at all. We introduce (i) a *refractory rotation* rule under which units that have just fired sit out the next competition, (ii) an *exact isolation* scheme that confines replay updates to synapses invisible to the current input under k-winner-take-all (k-WTA) dynamics, with per-synapse optimiser state advanced only inside the mask, and (iii) internal control signals — a unit-level homeostatic pressure that triggers replay bursts, and a relative-novelty (fast-over-slow error) gate that decides when rotation runs. On class-incremental split-MNIST the resulting system reaches $91.3 \pm 0.3\%$ with no offline phase, exceeding the best offline-night schedule ($90.7 \pm 0.4\%$), at a replay-sample cost of $\approx 1.2\times$ the night's; masked replay is insensitive to replay batch size down to eight samples, whereas unmasked replay and offline nights degrade sharply at the same sizes. The i.i.d. cost of rotation (1.3–3.0 points) is eliminated by the relative-novelty gate on learnable streams (95.0% vs. 95.0% without rotation) and reverses into a +2.5 to +6.5 point *gain* on harder data (split CIFAR-10), where the full ordering of methods transfers. We further show that surprise-triggered waking replay fails at any volume, that the trigger must be homeostatic, and that the residual advantage of offline nights is confined to very small episodic buffers on narrow substrates, and that sleep-anchored synaptic down-selection helps precisely where an offline window exists. We provide exactness proofs for the isolation rule and release all code, checkpoints and diagnostics.

1 Problem Formulation

Setting. Let a stream of labelled examples $(x_t, y_t) \in \mathbb{R}^d \times \{1, \dots, C\}$ be presented in T contiguous tasks $\mathcal{T}_1, \dots, \mathcal{T}_T$, where task $\mathcal{T}_\tau$ contains only classes $\mathcal{C}_\tau \subset \{1, \dots, C\}$, the $\mathcal{C}_\tau$ are disjoint, and no task identity is available at test time (class-incremental learning). A network f_θ with a single shared C-way readout is trained on the stream and evaluated on all C classes after the final task,

$$A_{\text{seq}}(\theta) \;=\; \Pr_{(x,y)\sim\mathcal{D}_{\text{test}}}\big[\arg\max_c f_\theta(x)_c = y\big], \qquad F \;=\; \frac{1}{T-1}\sum_{\tau=1}^{T-1}\big(a_{\tau,\tau} - a_{T,\tau}\big), \tag{1}$$

where $a_{t,\tau}$ is accuracy on task τ after learning task t and F is forgetting.

Constraints. We restrict the learner to a set $\mathcal{B}$ of cortical constraints: no weight transport (feedback synapses are learned, never copied), errors carried by a bounded burst-like channel, Dale's law, sparse population codes (k-WTA on wide layers), sparse distance-dependent connectivity, and a single forward–feedback sweep per example (no iterative relaxation). An episodic memory ("hippocampus") of at most K raw examples with reservoir writing is permitted; K is a budget, not a free parameter.

Objective. Unlike benchmark-chasing continual learning, we optimise a three-way trade-off

$$\max_{\theta} \left(A_{\text{seq}}, A_{\text{iid}}, -R, -E\right) \quad \text{subject to} \quad \theta \in \mathcal{B}, \tag{2}$$

where A_{iid} is accuracy of the identical learner on the i.i.d. shuffle of the same data (the *static axis*: a brain-like mechanism must not damage ordinary learning), R is the replay cost in *samples* $\times$ *synapses*, and E an event-driven energy proxy (Section 3). The central question of this paper: can the offline consolidation term be removed entirely — consolidation occurring only *during* inference on the wake stream — without losing A_{seq}, and at what cost in A_{iid} and R?

2 Related Work

Sleep-inspired consolidation in ANNs. The Bazhenov line implements sleep as a global offline phase: networks are converted to spiking dynamics and driven by noise under local Hebbian plasticity, recovering old tasks after each new one (Tadros et al., 2022), after several tasks at once (Bazhenov et al., 2026), in spiking networks with coarse alternation of training and sleep (Golden et al., 2022), and in equilibrium-propagation networks combined with small rehearsal buffers (Kubo et al., 2025). Wake–sleep frameworks with NREM and REM stages (Sorrenti et al., 2024; Robinson et al., 2022) and engineering systems such as SIESTA (Harun et al., 2023) likewise schedule consolidation offline. In all of these, sleep is global, offline, and clocked; none implements consolidation confined to currently unused units during ongoing inference.

Gating and subnetwork methods. Context-dependent gating (XdG) activates a fixed random subnetwork per task but requires task identity at train and test time (Masse et al., 2018); learned gates (Tilley et al., 2023), dendritic context vectors from input prototypes (Iyer et al., 2022), task-free sparsification by feedforward and top-down errors (Lässig et al., 2023), and gradient-recovered functional masks (McKee et al., 2026) remove the explicit label but all derive the gate from the *input identity.* The rotation studied here is instead a function of each unit's own *use history*, which, to our knowledge (literature search, August 2026), has no ANN precedent.

Complementary learning systems. CLS-ER maintains fast/slow EMA copies of the weights as semantic memories (Arani et al., 2022) and DualNet trains a slow self-supervised learner beside a fast supervised one (Pham et al., 2021); both raise the value of each replayed sample but still replay at every step. Our contribution is orthogonal: we change *where* (isolated synapses), *when* (homeostatic bursts) and at what granularity (micro-batches) replay is applied.

Pattern separation. Cerebellum-, mushroom-body- and dentate-gyrus-style sparse expansions (Cayco-Gajic et al., 2017; Cayco-Gajic and Silver, 2019; Litwin-Kumar et al., 2017) inspire continual learners such as the FlyModel (Shen et al., 2021), sparse distributed memory (Bricken et al., 2023) and DG-gated mixtures (Kapoor et al., 2026). We tested this family as a front-end and report a negative result (Section 5).

Biology of local sleep. Sleep-like off-periods occur locally in awake, sleep-deprived animals (Vyazovskiy et al., 2011). Causal evidence is recent: optogenetically inducing ON/OFF alternation in one cortical hemisphere of awake mice locally discharges sleep pressure, weakens synaptic markers (GluA1, pS845) and rescues memory consolidation under sleep deprivation, whereas an equal *tonic* reduction of firing does not (Driessen et al., 2026). Our refractory rotation is the algorithmic counterpart of this alternation requirement, and our silent-mask control the counterpart of tonic silencing.

3 Experiment Setup

Data and protocol. Sequential axis: split-MNIST (five tasks of two classes, 5 epochs per task) and split CIFAR-10 (same protocol; 32×32 luminance so that distance-dependent connectivity retains its geometry). Static axis: the identical learner and budgets on the i.i.d. shuffle. Waking batches have $n_w = 16$ samples; the episodic buffer holds $K \in \{200, 1000\}$ raw samples under reservoir writing. All numbers are mean ± s.d. over three seeds unless marked otherwise. A third regime uses a frozen V1-like feature front-end for CIFAR-10: 256 normalised random 6×6×3 patches sampled from the training images act as fixed filters (stride 2, rectification against the per-filter mean response, quadrant average pooling; 1024-D, linear probe ≈ 49%). Nothing in the front-end is learned; the cortex under study receives it as its retina.

Architecture. d–512–256–C with k-WTA sparsity 5% on the hidden layers (d–256–128–C at 10% where noted), 30% distance-dependent connectivity on hidden synapses, Dale's law with 80% excitatory units, and the single-sweep learning rule of Section 4.

Baselines. (i) Backpropagation with experience replay at equal buffer K (mandatory in every table); (ii) the offline *night*: after each waking epoch, 20 replay batches of 256 samples at 3× learning rate with no concurrent input (the strongest schedule from our earlier ablations); (iii) interleaved ER for the local learner; (iv) no buffer.

Metrics. Final accuracy, forgetting F, replay cost R (batches and samples), the isolated-channel width ρ_ℓ (fraction of replay-activated units in layer ℓ that are asleep for the current input), code overlap (mean pairwise Jaccard of task-active unit sets), participation-ratio dimensionality, and an event-driven energy proxy: 0.9 pJ per synaptic event for spiking inference vs. 4.6 pJ per MAC for dense rate inference, with thresholds calibrated post hoc and T_s integration steps (Merolla et al., 2014).

4 Methodology

4.1 Cortical base learner

Layer activities are $a^0 = x$ and, for $\ell = 1, \ldots, L$,

$$z^\ell = g^\ell \left(W^\ell a^{\ell-1}\right) + b^\ell, \qquad a^\ell = \Phi_k\big(\sigma(z^\ell) \odot (1 - s^\ell)\big), \tag{3}$$

where σ is a rectifier, $s^\ell \in \{0,1\}^{n_\ell}$ is the (possibly empty) *suppression mask* of Section 4.3, and Φ_k keeps the k largest entries of each row and zeroes the rest (k-WTA). Errors are produced by one top-down sweep: with target y,

$$\varepsilon^L = y - z^L, \qquad \varepsilon^\ell = \sigma'(z^\ell) \odot \Big(B^\ell \left[\kappa \tanh(\varepsilon^{\ell+1}/\kappa) \odot \mathbf{1}(a^{\ell+1} > 0)\right]\Big), \tag{4}$$

i.e. a bounded, event-gated burst signal carried by learned feedback weights B^ℓ (no transport). Forward and feedback weights follow the same local product with a shared decay λ (Kolen–Pollack alignment (Kolen and Pollack, 1994; Akrout et al., 2019)),

$$\Delta W^\ell \propto \varepsilon^\ell (a^{\ell-1})^\top - \lambda W^\ell, \qquad \Delta B^{\ell-1} \propto \varepsilon^\ell (a^{\ell-1})^\top - \lambda B^{\ell-1}, \tag{5}$$

with sign-concordant feedback under Dale's law and per-synapse adaptive scaling (ϵ-floored Adam). Equations (3)–(5) were fixed by earlier ablation ladders and are not varied here.

4.2 Exactly isolated replay

Let x be the current waking batch and define the *awake set* $\mathcal{A}^\ell(x) = \{i : a_i^\ell(x) \neq 0 \text{ for some sample in } x\}$. During the same forward step, a replay batch r drawn from the buffer is applied through the synapse mask

$$M_{ij}^\ell \;=\; \mathbf{1}\big[\, j \notin \mathcal{A}^{\ell-1}(x) \;\vee\; i \notin \mathcal{A}^\ell(x) \,\big], \tag{6}$$

i.e. a synapse may take the replay update iff its presynaptic unit is silent for the current input or its postsynaptic unit is asleep. Optimiser state (first and second moments) advances *only* inside the mask, so momentum cannot leak the update outside it. The readout row remains plastic (old classes must stay calibrated).

Proposition 1 (Pre-silent updates are exactly invisible). *Fix input x and let $\widehat{W}^\ell = W^\ell + \Delta^\ell$ with $\Delta_{ij}^\ell = 0$ whenever $a_j^{\ell-1}(x) \neq 0$. Then the network output on x is unchanged for any magnitude of Δ.*

Proof. By induction on ℓ. Assume $a^{\ell-1}$ is unchanged (true for $\ell = 1$ since $a^0 = x$). For any unit i, $\widehat{z}_i^\ell - z_i^\ell = g^\ell \sum_j \Delta_{ij}^\ell a_j^{\ell-1}(x)$. Each term vanishes: if $a_j^{\ell-1}(x) \neq 0$ then $\Delta_{ij}^\ell = 0$; otherwise $a_j^{\ell-1}(x) = 0$. Hence z^ℓ, and therefore a^ℓ through (3), is unchanged; the induction closes at the output layer. □

Proposition 2 (Post-asleep updates are invisible under a margin condition). *Let Δ^ℓ additionally allow $\Delta_{ij}^\ell \neq 0$ for units $i \notin \mathcal{A}^\ell(x)$ with active presynaptic partners. Write $\delta_i = g^\ell \sum_j \Delta_{ij}^\ell a_j^{\ell-1}(x)$ and let $z_{(k)}$ be the k-th order statistic of the post-nonlinearity values entering Φ_k. If for every such unit $\sigma(z_i^\ell + \delta_i) < z_{(k)}$ (the perturbed unit stays below the winner threshold), the output on x is unchanged.*

Proof. A unit outside $\mathcal{A}^\ell(x)$ contributes $a_i^\ell(x) = 0$. After the update its pre-activation becomes $z_i^\ell + \delta_i$; under the stated margin it is still excluded by Φ_k (or rectified to zero), so $a_i^\ell(x) = 0$ still, all other units' inputs are untouched by Proposition 1's argument, and the layer's output is identical. □

Corollary 1 (Suppressed units are exact for any magnitude). *If i is suppressed by the rotation mask ($s_i^\ell = 1$ in (3)), then $a_i^\ell(x) = 0$ identically and Proposition 2 holds with no margin condition.*

Remark 1. *Empirically the masked replay update changed the waking batch's output by* 0.0 *(double precision) in all logged checks; Corollary 1 explains why the refractory rotation below makes the isolation robust rather than merely approximate.*

4.3 Refractory rotation and its gates

Definition 1 (Refractory rotation). *After processing waking batch x_t, set $s_i^\ell(t{+}1) = \mathbf{1}[i \in \mathcal{A}^\ell(x_t)]$: every unit that fired sits out the next competition and is therefore asleep — and consolidable — for batch x_{t+1}.*

Rotation widens the isolated channel ρ_ℓ from 0.2–0.3 (natural silence) to ≈ 0.53, at the price of running inference on the second-best coalition. Two internal signals control the system (Figure 1):

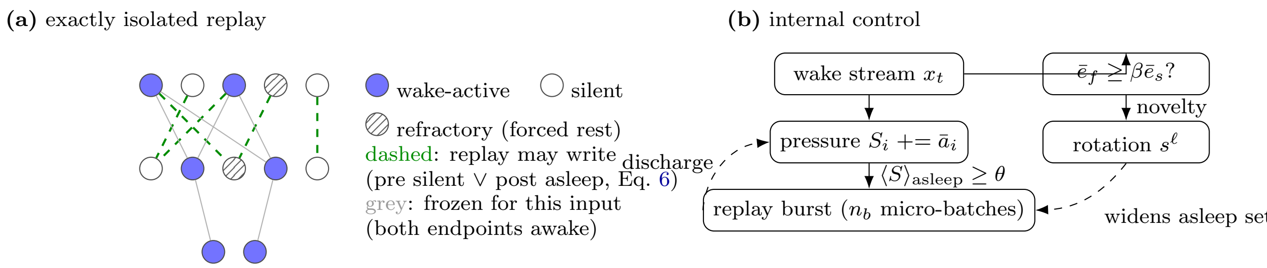

Figure 1: Mechanism. (a) During each waking batch, replay from the episodic buffer is written only into synapses whose update is provably invisible to the current input (Propositions 1–2); refractory rotation forces recently used units into the consolidable set (Corollary 1). (b) Two internal signals: unit-level homeostatic pressure triggers replay bursts; a relative-novelty gate decides when rotation runs.

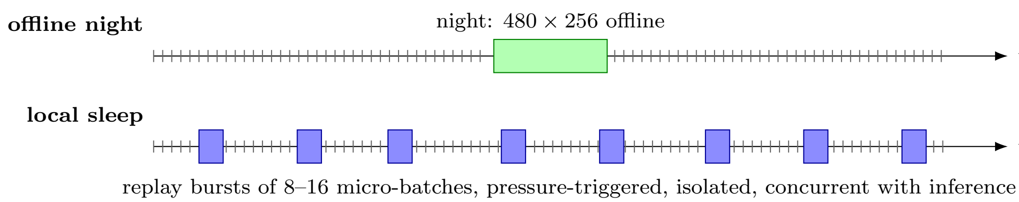

Figure 2: Consolidation schedules. Ticks are waking batches. The offline night pauses the stream; local sleep consolidates during it, in SWR-like bursts confined to asleep units.

Homeostatic pressure (when to replay). Each unit integrates its own use, $S_i \leftarrow S_i + \bar{a}_i(x_t)$, and a replay *burst* of n_b consecutive batches is triggered when the mean pressure of currently asleep units exceeds θ; consolidation discharges the pressure of the units that received it, $S_i \leftarrow (1 - \mathbf{1}[i \text{ asleep}])\, S_i$. This is a unit-level Process S (Borbély, 1982).

Relative novelty (when to rotate). With fast and slow EMAs of the batch error $\bar{e}_f(t) = (1-\alpha_f)\bar{e}_f + \alpha_f e_t$, α_f=0.1, and $\bar{e}_s$, α_s=0.002, rotation runs only while

$$\bar{e}_f(t) \;\geq\; \beta\, \bar{e}_s(t), \qquad \beta \approx 1.1\text{–}1.5, \tag{7}$$

i.e. while the stream is *new relative to the learner's own recent history* (an ACh-like adaptation signal (Hasselmo, 2006)). Absolute thresholds provably cannot serve here: the converged error of a ten-class i.i.d. stream exceeds any level a two-class task dips under, and we observe exactly this failure (gate open 100% of the time on static MNIST, 95% on CIFAR, for any threshold that behaves on split-MNIST). Because rotation is *beneficial* on streams the learner never masters (Section 5), the final gate adds an "unmastered" clause: with e_0 the mean error over the first 20 batches (a chance-level estimate), rotation runs while

$$\bar{e}_f(t) \;\geq\; \beta\, \bar{e}_s(t) \qquad \text{or} \qquad \bar{e}_f(t) \;\geq\; \gamma\, e_0, \qquad \gamma \approx 0.5. \tag{8}$$

The first clause opens the gate at task switches, the second keeps it open on any stream whose error remains a substantial fraction of chance — both scale-free.

4.4 Cost accounting

Replay cost is measured in samples (R_s = batches $\times$ batch size) and synaptic work. The night costs $R_s = 480 \times 256 \approx 1.2 \times 10^5$ samples. Continuous local sleep with micro-batches of 8 costs

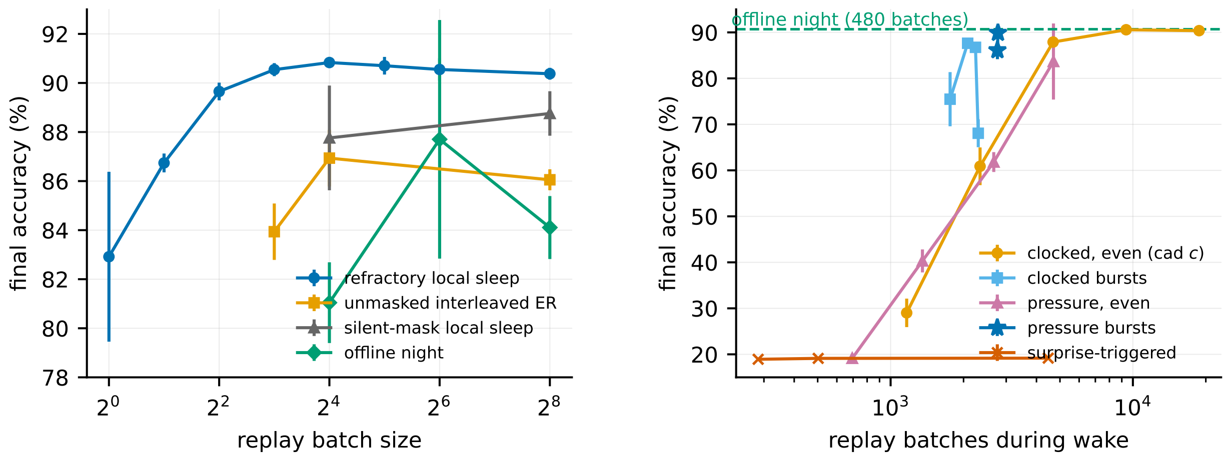

Figure 3: Left (isolation): final split-MNIST accuracy vs. replay batch size. Masked refractory replay is insensitive down to 8 samples; unmasked, silent-masked and offline-night replay all degrade. **Right (timing):** accuracy vs. number of waking replay batches. Even trickles and rare naps fail; bursts of 8–16 succeed; unit-level pressure triggering (stars) dominates at equal budget; surprise triggering fails at any volume.

18,760 $\times$ 8 $\approx$ 1.5 $\times$ 10^5 samples (1.2$\times$ the night); pressure-triggered bursts with batch 64 cost 1.8 $\times$ 10^5 (1.4$\times$).

5 Results

5.1 Local sleep replaces the night

On split-MNIST at K=1000, refractory local sleep with replay every other waking batch (batch 64) reaches $\mathbf{91.3 \pm 0.3}$% with no offline phase, above the best offline night (90.7 ± 0.4%) and far above unmasked interleaved ER at the same budget (86.1–86.9%). Adding a night on top of continuous local sleep yields nothing (90.9 ± 0.3 vs. 90.8 ± 0.1): once consolidation runs during wake, the offline phase is redundant at this buffer size.

5.2 Isolation stabilises micro-batch replay

Figure 3: masked refractory replay is flat in replay batch size from 256 down to 8 samples ($90.4 \to 90.5$%), while unmasked ER falls to 83.9 ± 1.2, the silent mask to 87.8 ± 2.1, and the offline night to 81.0 ± 1.6 at batch 16. Isolation — including optimiser state confined to the mask — is what makes replay affordable: the earlier "39×" cost of inference-time consolidation was an artefact of counting 256-sample batches.

5.3 When replay must be rare: homeostatic bursts, never surprise

Figure 3 (right): at ∼12% of replay events, clocked bursts of 8–16 batches retain 86.8–87.6% while an even trickle collapses (60.9 ± 4.1) and rare long naps also fail (63–70%) — masked replay reaches only ∼half the synapses per event and can neither wait nor dribble. Pressure-triggered bursts reach $\mathbf{89.9 \pm 0.1}$% at 15% of events (2,773 batches; 1.4× night samples with batch 64). Surprise-triggered waking replay fails at *any* volume ($\approx$ 19% even at 4,472 events): novelty fires only after task switches

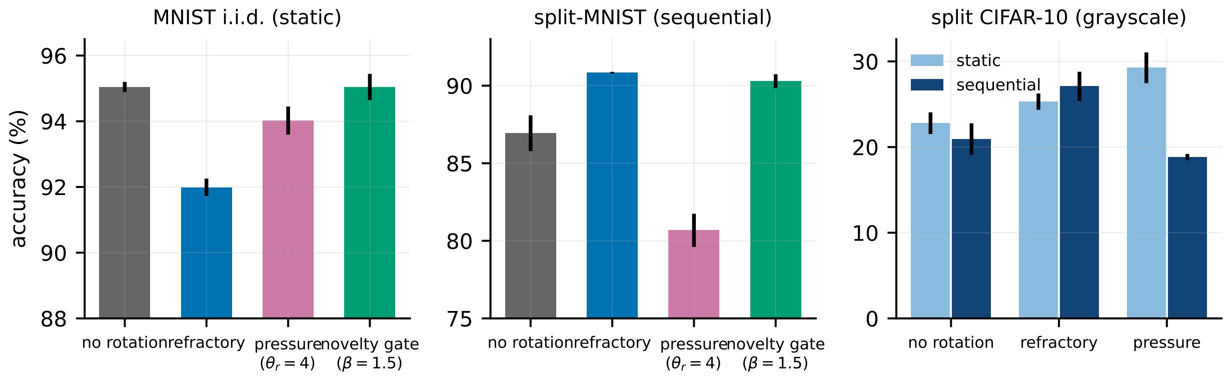

Figure 4: Rotation: price, trade-off, and gate. Left: on i.i.d. MNIST the novelty gate restores no-rotation accuracy. Middle: on split-MNIST rotation is required and the gate keeps it. Right: on split CIFAR-10 (grayscale) rotation is a gain on *both* axes.

and starves later epochs. The waking trigger must be homeostatic — the opposite of the night's optimal trigger, which our earlier controller showed *should* be novelty-timed.

5.4 The i.i.d. price of rotation and the novelty gate

Figure 4: always-on rotation costs 1.3–3.0 points on i.i.d. MNIST (95.0 → 92.0–93.7). Gating rotation by *unit-level pressure* only trades the axes (static 91.6 → 94.0 as sequential 88.8 → 80.7; no dominant point). The *relative novelty* gate (7) removes the price at no sequential cost on MNIST: static $\mathbf{95.0} \pm \mathbf{0.4}$ (rotation on 1.4–17% of batches), sequential 90.3 ± 0.4 (β=1.5) or 90.9 ± 0.6 (β=1.1). On split CIFAR-10 the price *reverses*: rotation gains +2.5 (refractory) to +6.5 points (pressure-selected rest) on the static axis — forced rotation acts as a regulariser in the underfit regime — while the pure novelty gate (7) *misfires* there (sequential 19.1 ± 1.1: fast/slow → 1 on a stream that never becomes predictable, so rotation shuts off exactly where it helps). The progress-keyed gate (8) resolves this with one setting across all four regimes: split CIFAR-10 sequential 27.2 ± 0.7 and static 25.3 ± 1.0 (both = always-on rotation), split-MNIST 90.4 ± 0.9, i.i.d. MNIST 94.3 ± 0.2 at γ=0.5 — retaining a residual 0.7-point static cost relative to the pure novelty gate on easy streams, the price of the unmastered clause.

5.5 Transfer to CIFAR-10

On split CIFAR-10 (grayscale, same protocol) the ordering transfers and widens: refractory local sleep 27.1 ± 1.7 > BP+ER 25.1 ± 1.5 > offline night 23.8 ± 1.2 > unmasked ER 20.9 ± 1.8 > no buffer 16.0–16.5, despite task-active code overlap of 0.9–0.98. On the V1-like feature front-end the ordering among local variants persists and widens (refractory 37.0 ± 0.4 > progress-gated 36.0 ± 0.6 > offline night $31.7 \pm 1.5 \approx$ unmasked $30.8 \pm 0.6 \gg$ no buffer 17.0; static rotation gain +3.9: 35.6 ± 1.3 vs. 31.7 ± 3.6), but BP+ER reaches 42.6 ± 0.5 and overtakes the local learner — the reverse of the raw-pixel result. The local-sleep advantage over backpropagation with replay is therefore regime-specific (local rules, sparse connectivity, raw inputs), while the advantage over *other local schedules* (nights, unmasked ER) is robust across all three input regimes.

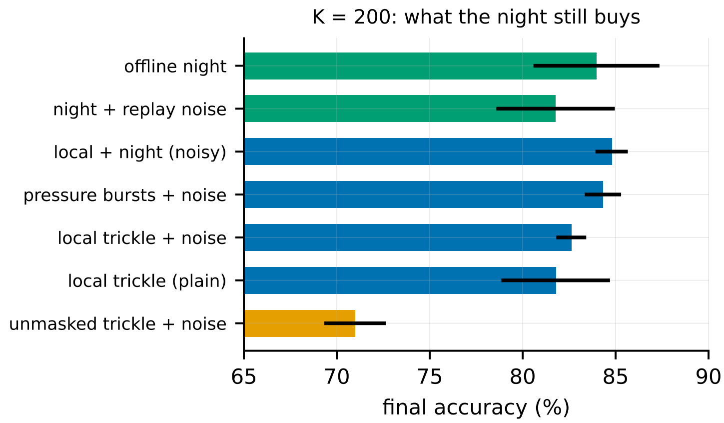

Figure 5: $K = 200$ **mechanism triangulation.** Isolation helps, diversity helps only during wake, and the irreducible advantage of the night at tiny buffers is its interference-free window.

5.6 Small buffers: what the night still buys

At K=200 the night appeared to keep a ∼2.6-point edge in our earlier phases; the final triangulation (Figure 5) dissolves most of it into a substrate mismatch. On the *same* 512–256/5% substrate, the night reaches 84.0 ± 3.4 while noisy pressure-burst local sleep reaches 84.3 ± 1.0 and local+night 84.8 ± 0.9 — parity, with three-fold smaller variance for the wake-side scheme. The strongest tiny-buffer number remains the narrower 256–128/10% night (86.9), so the offline window retains an advantage only in combination with narrow substrates. The mechanism results stand: replay diversity (noise σ=0.2) is specifically a wake-side lever — it lifts every local scheme ($81.8 \pm 2.9 \rightarrow 84.3 \pm 1.0$) and *hurts* the night ($84.0 \rightarrow 81.8 \pm 3.2$) — and *unmasked* waking replay collapses (71.0 ± 1.7): isolation is a benefit, not a cost, at every buffer size we tested.

5.7 Energy

Converted to spiking inference (T_s=8), the full system runs at 91.6% and 119 nJ/sample vs. 2461 nJ for dense rate inference (20.7×) and 516 nJ for event-driven rate inference (4.3×); the rotation-trained network loses accuracy beyond T_s=8 (90.4 at 16), unlike its non-rotated control (94.1) — an open interaction between rotation-shaped weights and threshold calibration.

5.8 Sleep-anchored down-selection

Motivated by the synaptic homeostasis hypothesis (Tononi and Cirelli, 2014), we anchored structural plasticity (prune the weakest fraction of live synapses, regrow the same number with a distance bias) to consolidation windows, using the Kolen–Pollack decay of (5) as the natural weakening force on unreplayed synapses. The verdict splits three ways at matched turnover (≈ 5% per epoch). On the *continuous* local-sleep substrate pruning is free (sleep-anchored 90.9 ± 0.2, clocked 90.6 ± 0.4, none 90.8 ± 0.1). On the pure *night* substrate, pruning immediately after each night helps substantially (87.2 ± 3.8 vs. 84.1 ± 1.3) — offline down-selection rescues the wide-sparse night. On the *episodic-burst* substrate pruning is harmful (84.6 ± 0.5 burst-anchored, 86.6 ± 1.3 clocked, vs. 89.9 ± 0.1 without): when replay is rare, synapses are cut before enough consolidation events have protected them. Down-selection, like replay itself, needs either density or an offline window.

5.9 Negative results

For the record, the following were falsified under matched budgets: a frozen dentate-gyrus-style sparse expansion front-end (input decorrelation does not propagate through learned k-WTA layers: task overlap at the input drops $0.79 \to 0.18$ yet every downstream metric worsens by 3–11 points); generative (REM) replay as a substitute for episodic samples; margin-based reverse learning; surprise-gated memory *writing*; multiplicative burst coding; absolute-threshold rotation gates; pruning anchored to sparse replay bursts (above).

6 Discussion

Why isolation stabilises micro-batches. A replay micro-batch is a high-variance gradient estimate. Applied to the full network it perturbs the very coalition currently encoding the wake stream, and with shared optimiser state the perturbation compounds (cf. the unmasked curve of Figure 3). Inside the mask, each update is provably output-neutral for the present input (Propositions 1–2), so variance accumulates only where it can do no immediate harm and is averaged over thousands of events. The offline night enjoys no such protection: with nothing clamping the network, a small noisy batch at 3× learning rate moves the whole weight state.

Correspondence with biology. The refractory rule mirrors the causal finding of Driessen et al. (2026) that ON/OFF *alternation*, not tonic silencing, discharges local sleep pressure and rescues memory — our silent-mask control (their tonic condition) is reliably worse. The dissociation of triggers — homeostatic for waking consolidation, novelty-timed for full nights — is a testable prediction for biology. The regime dependence of the rotation price (cost on easy streams, gain under chronic underfitting) may bear on why local sleep in vivo both impairs behaviour (Vyazovskiy et al., 2011) and serves useful functions when appropriately timed.

Honest limits. All results are at MLP scale on MNIST/CIFAR variants with three seeds; BP+ER overtakes the local learner on feature-based CIFAR, bounding the regime of our headline comparison; the progress gate keeps a 0.7-point static residue; and the novelty claims rest on our own literature search (August 2026).

7 Future Work

(i) Explaining why backpropagation with replay overtakes the local learner specifically on feature representations (dense connectivity, per-synapse adaptivity, or the learning rule itself); (ii) removing the residual 0.7-point static cost of the progress-keyed gate; (iii) the narrow-vs-wide substrate interaction of offline nights at tiny buffers; (iv) convolutional and deeper stacks; (v) a throughput theory of the isolated channel predicting the required replay cadence from ρ_ℓ, code overlap and the interference rate; (vi) the rotation–spiking calibration interaction beyond T_s=8.

8 Conclusion

Consolidation does not require an offline phase. With exactly isolated replay, refractory rotation of the active coalition, homeostatic burst triggering and a relative-novelty gate on the rotation itself, a biologically constrained learner consolidates *while it infers*: it matches and slightly exceeds the strongest offline-night schedule on split-MNIST at ∼1.2× the night's replay samples, pays nothing on learnable i.i.d. streams, gains outright on harder data, and runs at a twentieth of dense-rate inference energy when spiking. The offline night retains one irreducible role — consolidating very small episodic stores — which sharpens, rather than blurs, the division of labour between wake and sleep.

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